



Utilizing Hyperspectral Satellite Data for Geological Mapping and Mineral Exploration

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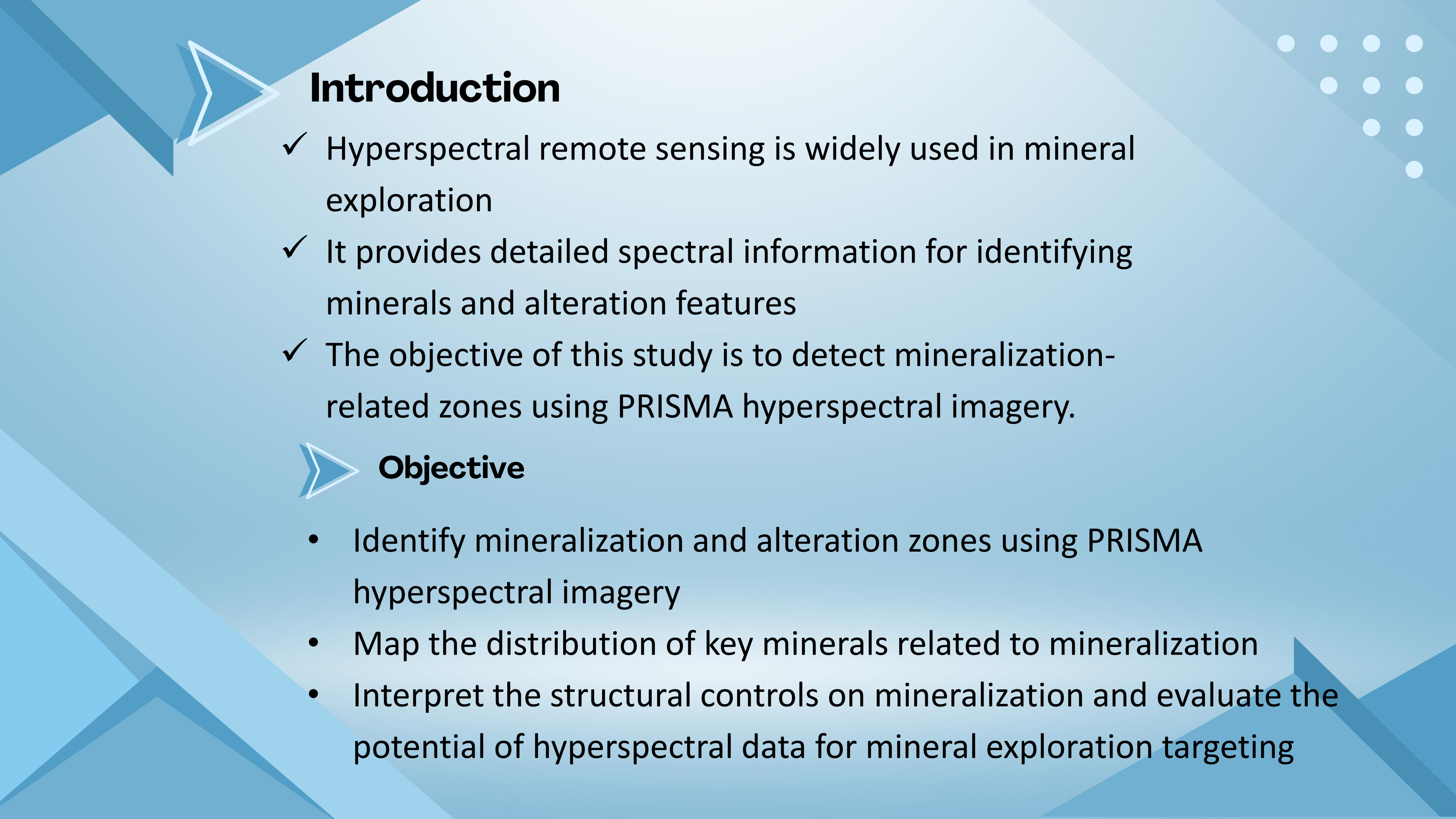
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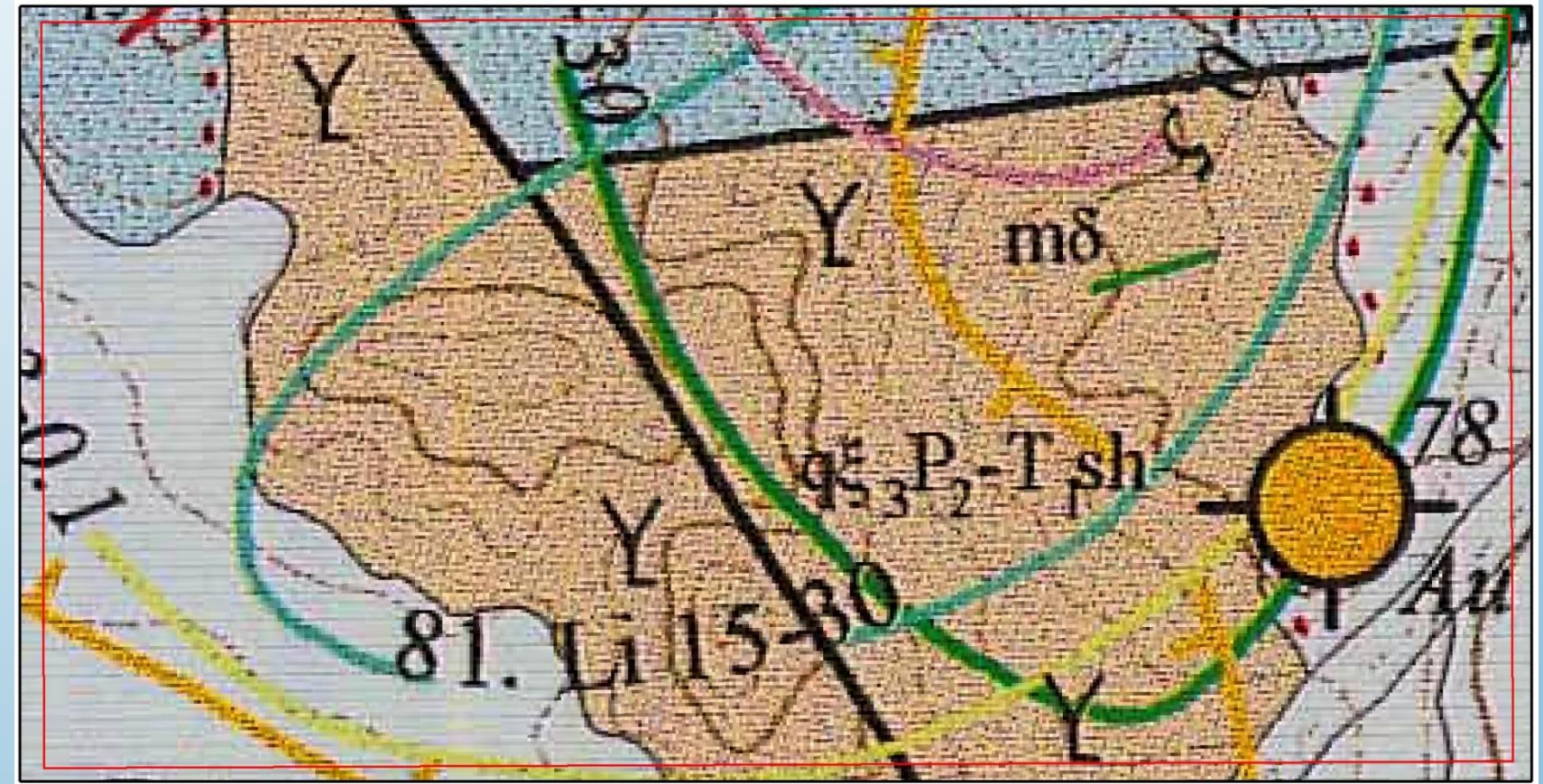
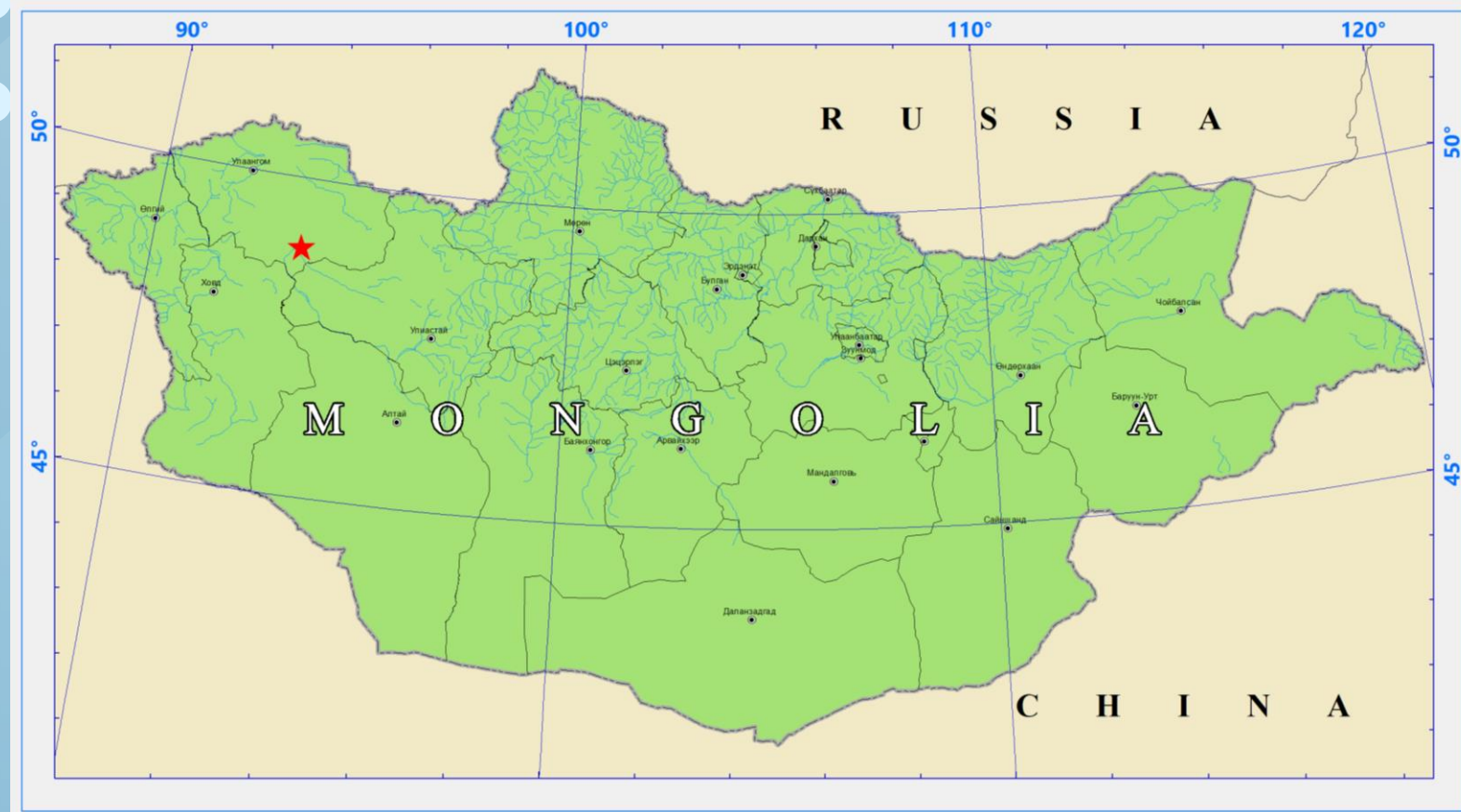
Introduction

- ✓ Hyperspectral remote sensing is widely used in mineral exploration
- ✓ It provides detailed spectral information for identifying minerals and alteration features
- ✓ The objective of this study is to detect mineralization-related zones using PRISMA hyperspectral imagery.

Objective

- Identify mineralization and alteration zones using PRISMA hyperspectral imagery
- Map the distribution of key minerals related to mineralization
- Interpret the structural controls on mineralization and evaluate the potential of hyperspectral data for mineral exploration targeting

Study area



The study area is located in Zavkhan Province, western Mongolia. The study area covers approximately 3.85 km², representing a focused target zone for hyperspectral mineral mapping.

Geologically, the area is underlain by intrusive rocks of granitoid and intermediate composition.

The central part is dominated by syenite intrusions, the eastern sector by diorite rocks, and the northern portion by andesitic volcanic rocks. The western part is covered by Quaternary deposits.

Importantly, the area contains structural lineaments and fault zones that cut through the intrusive body. These structures control hydrothermal fluid flow and mineralization.



Data methodology

1. Spectral Angle Mapper (SAM)

- Comparison with reference spectra
- USGS Spectral Library used as spectral reference
- Mineral classification based on spectral similarity

2. Band Ratio Analysis

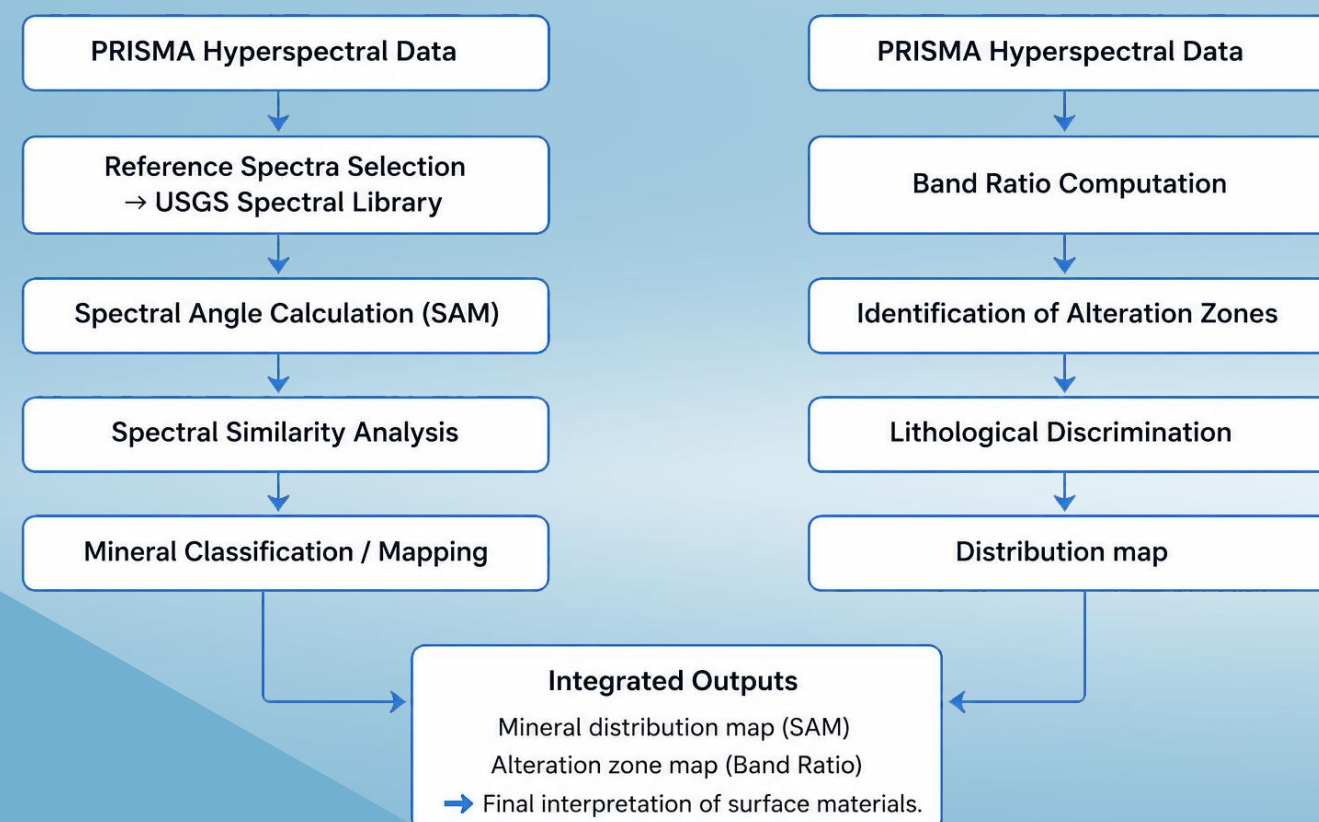
- Enhancement of diagnostic absorption features
- Identification of alteration zones
- Lithological discrimination

PRISMA

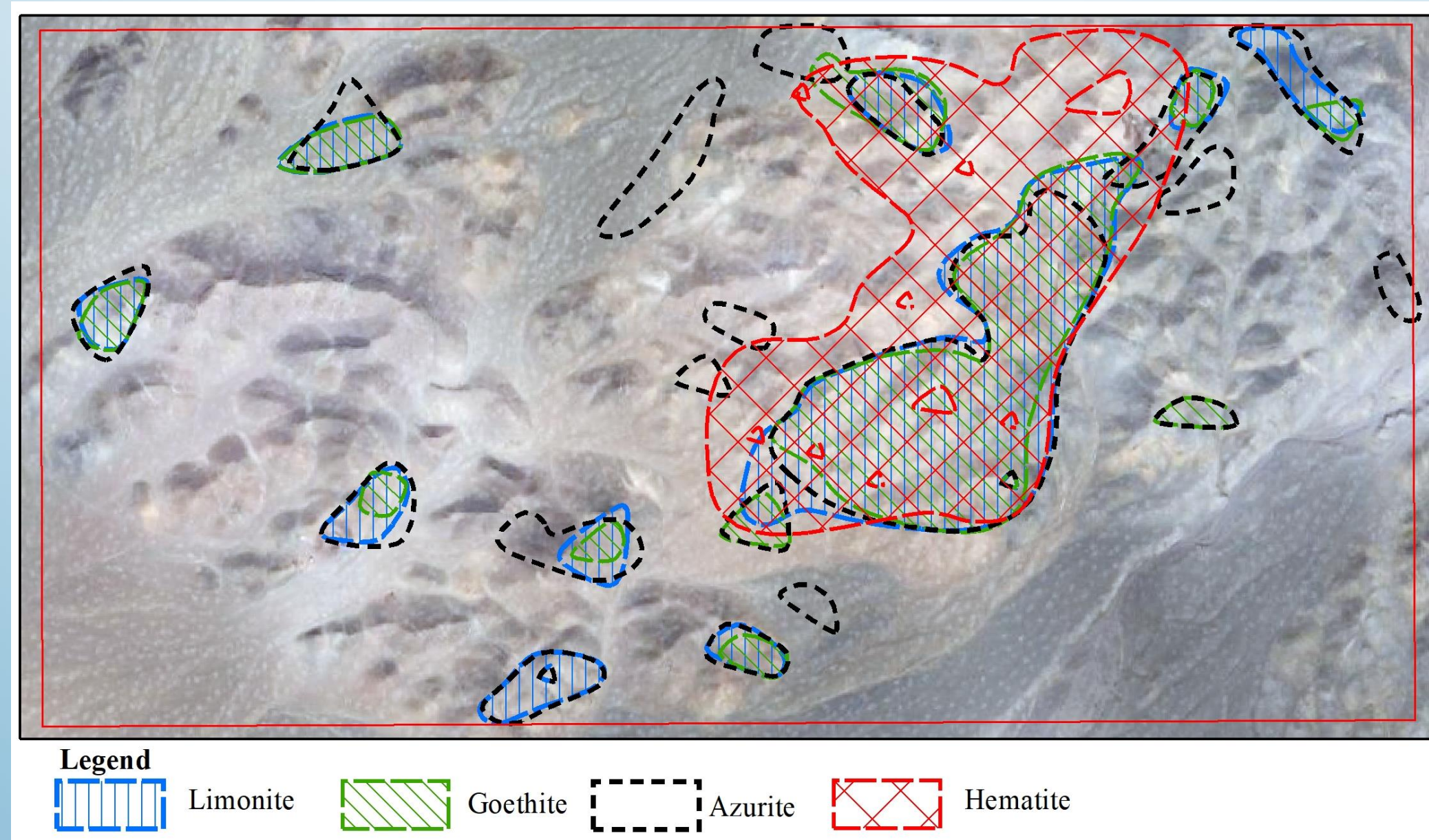
- PRISMA hyperspectral imagery
- Spatial resolution: ~30 m
- Spectral resolution: ~12 nm
- Suitable for:

Mineral exploration
Geological mapping
Environmental monitoring
Lithological discrimination

Hyperspectral Data Analysis Workflow (PRISMA)



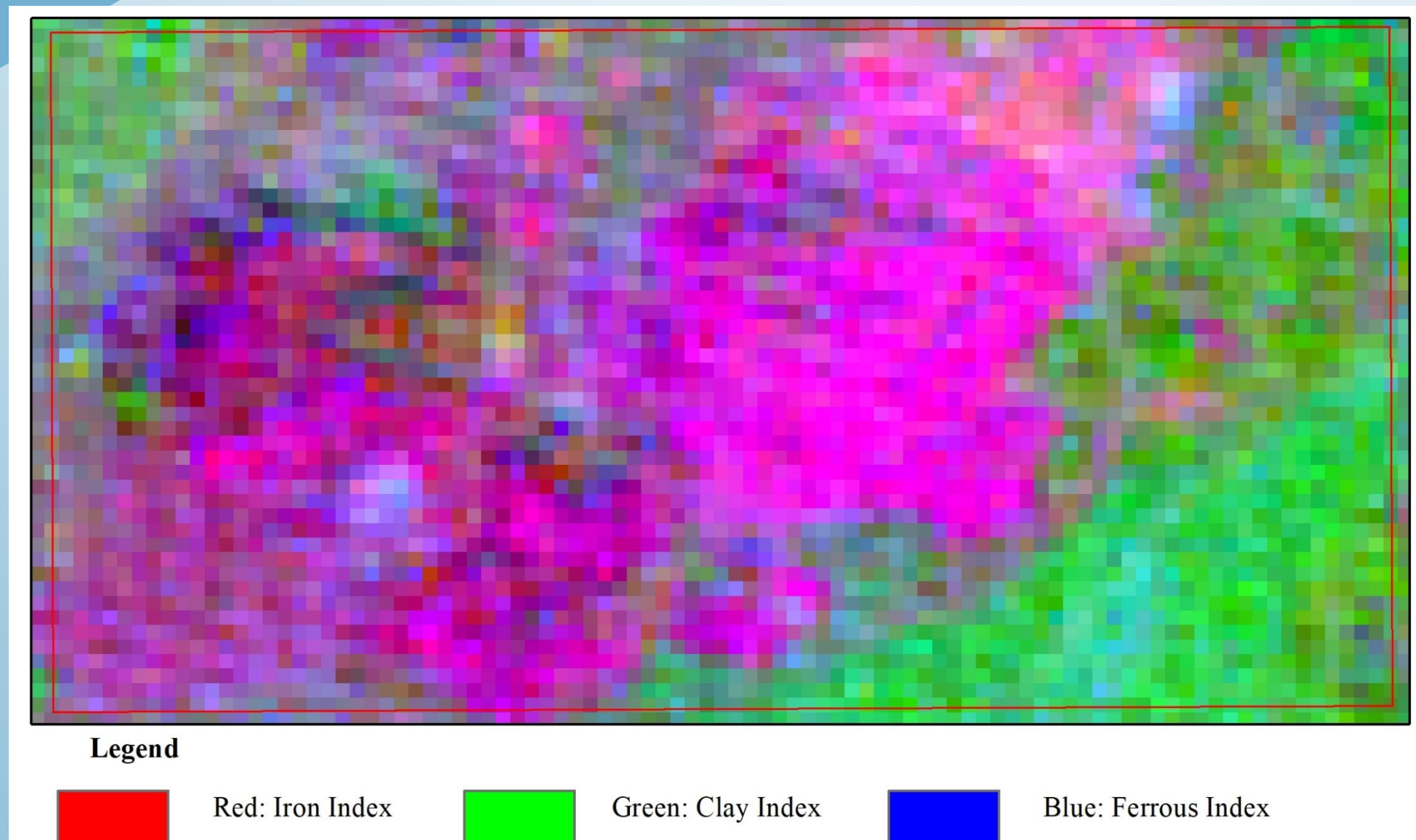
Results



Spatial Association of Iron Oxides and Azurite

The central part of the study area shows a strong spatial overlap of limonite, goethite, hematite, and azurite. This association indicates a well-developed oxidation zone formed by weathering processes. The presence of iron oxides together with secondary copper minerals suggests a possible gossan-related hydrothermal alteration zone. Although primary sulfide minerals were not directly detected in the same locations, these alteration patterns may represent surface expressions of deeper mineralization and should be verified through field investigation and geochemical analysis.

Results



Alteration map

The alteration map was derived using Band Ratio Analysis applied to PRISMA hyperspectral imagery. Bright magenta and pink tones indicate zones of intense hydrothermal alteration, including argillic alteration assemblages. Green regions represent less-altered or fresh rock units. The spatial clustering of altered zones closely corresponds to known mineralization targets at the area, confirming the effectiveness of band ratio techniques in delineating alteration halos associated with potential mineralization zones.

Band Ratio Composite

- **Iron Index (Fe^{3+})=Red Channel (Red / Blue visible ratio):**

Visible Red (Red band ~ 630 nm) / Visible Blue (~ 450 nm)

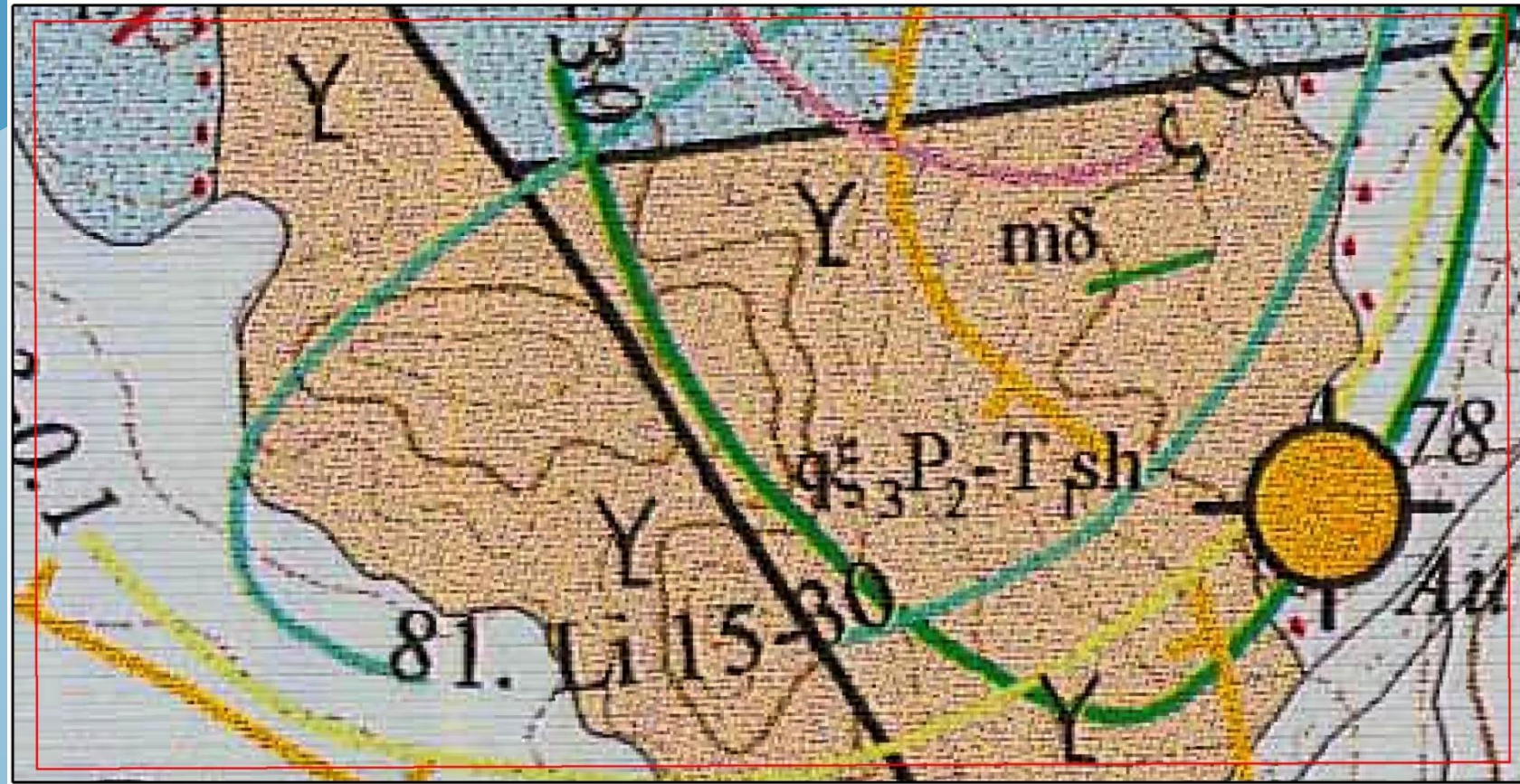
- **Clay Index=Green Channel (SWIR ratio):**

Shortwave Infrared (SWIR ~ 2200 nm) / SWIR ~ 2100 nm

- **Ferrous Index (Fe^{2+})=Blue Channel (NIR ratio):**

Near Infrared (NIR ~ 1000 nm) / Near Infrared (~ 900 nm)

Results



Geological map



Lithological map (Spectral Index RGB Composite)

The RGB composite is constructed using weighted spectral indices to enhance lithological and hydrothermal alteration discrimination. The red channel represents the iron-related index (Fe^{3+} / iron oxide), emphasizing oxidized and weathered zones. The green channel corresponds to the clay mineral index (Al–OH absorption feature in the SWIR region), highlighting hydroxyl-bearing alteration minerals such as kaolinite and illite. The blue channel represents the ferrous iron (Fe^{2+}) index derived from VNIR spectral response, associated with primary mafic minerals such as pyroxene. Scaling factors of $R = 1.6$, $G = 2.2$, and $B = 0.65$ were empirically determined to optimize spectral contrast and improve feature separation. The composite reveals distinct spectral zones: dark tones indicate intense Fe^{3+} oxidation and gossan development; purple to violet zones reflect argillic alteration; pinkish hues correspond to kaolinite- and illite-rich assemblages; and green tones represent less-altered mafic lithologies. The resulting composite effectively delineates lithological units and highlights potential mineralized and hydrothermally altered zones.

Conclusions

- ❖ PRISMA hyperspectral data, combined with SAM and Band Ratio techniques, successfully mapped mineralization indicators and hydrothermal alteration zones across the study area.
- ❖ Spatial patterns of mineral occurrences indicate that mineralization is strongly controlled by geological structures, particularly faults, fractures, and intrusive contacts.
- ❖ The association of limonite, goethite, hematite, and azurite suggests the presence of extensive oxidation zones and possible gossan-related alteration areas.
- ❖ Hyperspectral-derived mineral and alteration maps provide valuable indicators for interpreting mineralization processes, suggesting a structurally-controlled hydrothermal origin likely associated with Permian–Triassic intrusive activity
- ❖ Given Mongolia's vast and remote terrain, hyperspectral remote sensing offers a cost-effective and time-efficient approach for large-scale mineral exploration, significantly reducing the need for extensive fieldwork in the early stages of exploration.
- ❖ Integration of hyperspectral analysis with geological mapping, geochemical data, and field verification is essential for reliable exploration targeting and mineral potential evaluation.

References

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***Thank you for your
attention***